

Automated SOAP Note Generation in Speech-Language Pathology: A Multi-Agent Framework with Prompt Optimization

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Abstract

A SOAP Note is a clinical documentation format used to summarize doctor–patient interactions across four sections: Subjective, Objective, Assessment, and Plan. While effective for record-keeping, manual preparation increases clinician workload, especially in speech-language pathology where therapy and Note-taking must occur simultaneously. This report presents a multi-agent system for automating SOAP Note generation, incorporating agents for Note creation, interaction classification, and presence verification. To enhance performance, TextGrad-based prompt optimization was applied, with GPT-4o and Qwen3(Yang et al., 2025) evaluated on real doctor–patient conversations and corresponding clinician-authored Notes. Results show significant improvements in capturing clinically relevant details, demonstrating the potential of this approach to reduce documentation burden while ensuring improvement in quality and evaluation.

1 Introduction

With the widespread adoption of large language models, there has been a lot of interest in employing them to automate domain-specific tasks in the field of healthcare, legal and several other fields. A SOAP Note is a clinical documentation format used to summarize doctor–patient interactions. The acronym stands for Subjective, Objective, Assessment, and Plan, representing the key components of the conversation. Clinicians typically use this structure to capture the essential details of each session, making it easier to review past discussions and develop an appropriate plan for the following week. However, manually preparing SOAP Notes reduces efficiency and increases the workload of the clinicians. Since doctor-patient conversations are mostly set in the field of speech-language pathology, clinicians find it difficult to conduct therapy and take notes in real

time. The difficulties in this approach usually lie in model hallucinations, lack of automated evaluation to evaluate the generations and the generated SOAP Note, not matching the style of clinicians, especially in the Assessment and Plan sections, where the goal is to generate new insights based on the conversation.

This research project focuses on addressing these challenges by experimenting with different agentic frameworks and evaluating the suitable one according to domain-specific metrics. There are two primary research questions:

1. Given the scope of this study, How can SOAP Notes be generated to better align with clinician style and quality?
2. How can the quality of generated Notes be evaluated to save time and support iterative improvement?

To answer these questions, this study makes the following contributions:

- Different multi-agent approaches combined with automated agentic evaluation to improve and evaluate the quality of the generated SOAP Note.
- Optimized prompt, inspired by Wu et al. (2025), which helps to enhance generation to be closer to the style of clinicians, without requiring extra compute.

2 Related Works

2.1 Previous Approaches

Several approaches have been explored with respect to automating the generation of SOAP Notes with LLM-based methods. Brake and Schaaf (2024) compares two approaches, mainly a single model for generating the entire SOAP Note and multiple models which target different

sections of the SOAP Note. According to this paper, the approach of multiple models has the drawback of not being able to demonstrate overall coherence or consistency between the different sections since they are separately generated. In our future work, we hope to explore the difference in quality of generation between a single agent and a multi-agent system to address these gaps.

Li et al. (2025) introduce a novel framework for generating high-quality SOAP Notes by extracting salient keywords from doctor–patient interactions and incorporating these keywords into the prompting process. Notably, however, the conversations examined in their study are predominantly limited to routine check-up scenarios. In contrast, our work extends beyond such settings by engaging with more diverse, complex, and representative real-world doctor–patient dialogues.

Davis et al. (2025) present a framework for extracting relevant sections from clinical Notes and structuring them into a SOAP Note format: most notably the History of Present Illness, Interval History, and Assessment and Plan. Their method employs fuzzy matching to identify and extract the appropriate start and end points for each section. While this represents a noteworthy approach, it differs from our work in that it focuses on extraction from clinician-authored Notes rather than directly from clinician–patient conversations.

Chen and Hirschberg (2024) propose two approaches for summarizing doctor–patient conversations: a general summarization model and a SOAP-oriented model that produces summaries in SOAP format. Their study highlights the limitations of these approaches, such as the hallucination of medical terms when applied to out-of-domain data. Building on this, our work aims to mitigate these challenges and further enhance performance, by employing an agentic AI approach.

2.2 Application of AI Agents in Healthcare

AI agents have previously been explored for automating various healthcare processes. Lu et al. (2024) introduce a novel multi-agent framework that automates case classification in hospital Emergency Departments based on the Emergency Severity Index. Their approach leverages multiple agents engaged in role-based interactions, integrating

mechanisms such as self-confidence estimation and early stopping to achieve accurate assessments of case severity.

2.3 Prompt Optimization in Healthcare

Since the quality of prompts plays a role in the quality of the generations, a lot of research has been conducted in the field of prompt optimization. One such paper, Wu et al. (2025) employs Textgrad (Yuksekgonul et al., 2024) in its framework to optimize the prompt to increase the accuracy in the field of medical question and answering. In our paper, we have employed this in our system prompt to enable the agent to generate a SOAP Note much closer to the ground truth that we currently have.

3 Data

The data used for this project are the real-world transcripts of doctor–patient conversations, prepared by Prof. Yao Du and her fellow clinicians by interviewing four different patients, who will be called M.A., R.T., L.V.G. and F.W. for the sake of anonymity. We obtained corresponding ground truth SOAP Notes for a majority of these conversations, which enabled us to get a better idea about the kind of SOAP Note that clinicians expect. There are 9 M.A. conversations and 8 R.T. conversations which have corresponding ground truth SOAP Notes that have enabled us to research the gaps in our generations and how to improve them.

4 Previous Experimented Approaches and their Limitations

4.1 Single Agent Generation and Multi-Agent (SO+AP) Approach

This approach as demonstrated in Fig. 1, involves a comparison of two approaches, a single agent generating the entire SOAP Note and multiple agents splitting the generation of the SOAP Note between them, in this case, SOAP Note is generated by two agents which are split into Agent(S+O) and Agent(A+P).

These are then evaluated by an evaluator agent which takes a ground truth SOAP Note as input and rates the SOAP Note from 1-10, based on overall coherence and consistency. The best of two pairs is then passed to a sub-workflow, which consists of the same evaluator and a refiner agent. The SOAP Note is looped between the evaluator and the refiner, until it reaches a score of 9/10 or the maximum iterations of 10 is complete.

Limitations

- Evaluation is done by an LLM Agent, which is not trustworthy. Scores generated by the evaluation agent have a high chance of being probabilistic.
- While the multi-agent approach does not show inconsistency, it also doesn't reflect what an actual real-world SOAP Note looks like.

4.2 Single Agent Generation and Multi-Agent (S+O+AP) Approach

This approach as demonstrated in Fig. 2, involves a comparison of two approaches, a single agent generating the entire SOAP Note and multiple agents splitting the generation of the SOAP Note between them, in this case, SOAP Note is generated by three agents which are split into Subjective Agent(S), Objective Agent(O) and Assessment and Planning Agent(A+P).

These are then evaluated by an evaluator agent which takes a ground truth SOAP Note as input and rates the SOAP Note from 1-10, based on overall coherence and consistency. The best of two pairs is then passed to a sub-workflow, which consists of the same evaluator and a refiner agent. The SOAP Note is looped between the evaluator and the refiner, until it reaches a score of 9/10 or the maximum iterations of 10 is complete.

Advantages

- Follows the structure of the rubric at each section of the SOAP Note, better than the single agent approach.
- Gives a lot more detailed summarization of the doctor-patient conversation and refines the SOAP Note further.

Limitations

- Evaluation is done by an LLM Agent on criteria which were not verified by clinicians, making it untrustworthy. Scores generated by the evaluation agent have a high chance of being probabilistic.
- While the multi-agent approach does not show inconsistency, it also doesn't reflect what an actual real-world SOAP Note looks like.

5 Current Methodology

5.1 Overall Framework Architecture

The overall framework mainly involves three agents:-

1. A SOAP Note agent which generates the SOAP Note of the corresponding doctor-patient conversation in the format of four sections:- Subjective, Objective, Assessment and Plan.
2. A communication breakdown classification agent which, due to the clinician style in the Objective section of the ground truth SOAP has been extended to detect both breakdowns and successful instances of conversations.
3. A presence verification agent which checks the presence of these communication instances in the Objective section of the SOAP Note to evaluate the amount of information captured by the SOAP Note, to a certain extent.

The doctor-patient conversations are passed to the SOAP Note agent and the classification agent, which generate the SOAP Note and the patient-AI interactions from the corresponding conversation respectively. These outputs are then passed into the presence verification agent which counts the number of interactions present in the generated SOAP Note.

5.2 Prompt Optimization Pipeline

5.2.1 Prompt Optimization for Classification Agent

The prompt optimization for the classification agent involved having to take the conversations and the corresponding SOAP Notes from the recent data which contained patient-AI interactions. The objective section of these SOAP Notes were extracted and placed in a dictionary with the respective conversations. These were then split into train and test data with a 60:40 split. The training data was used to optimize the prompt and the test data was used to evaluate the quality.

5.2.2 Prompt Optimization for SOAP Note generation Agent

The prompt optimization for the SOAP Note agent involved having to take all the conversations and their corresponding SOAP Notes. The SOAP Notes and their corresponding conversations were placed

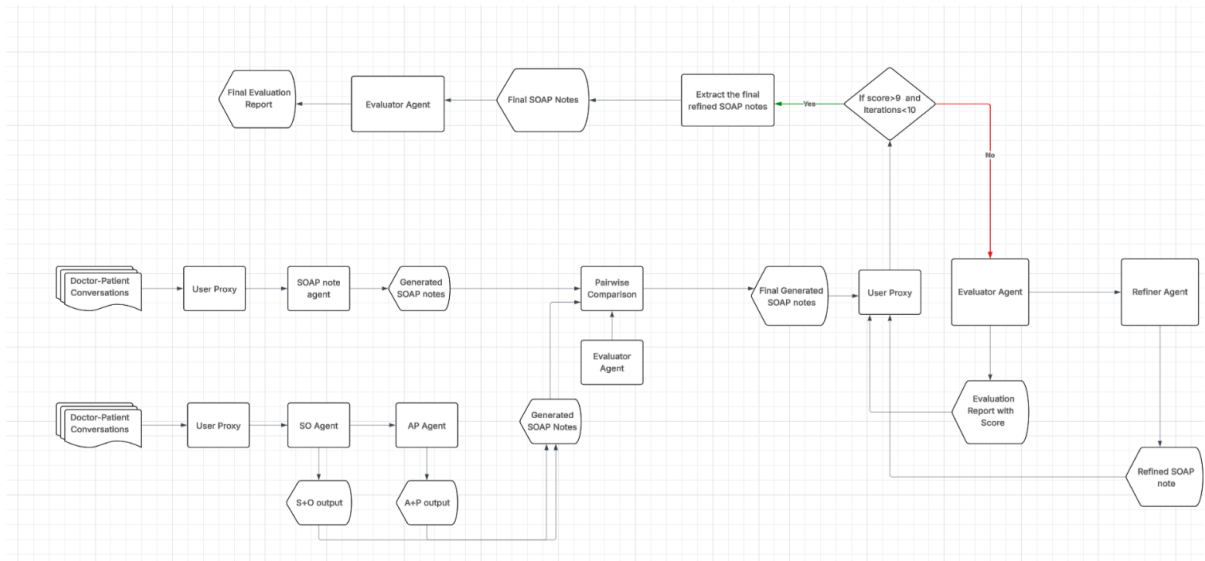


Figure 1: Single Agent Generation and Multi-Agent (SO+AP) Approach

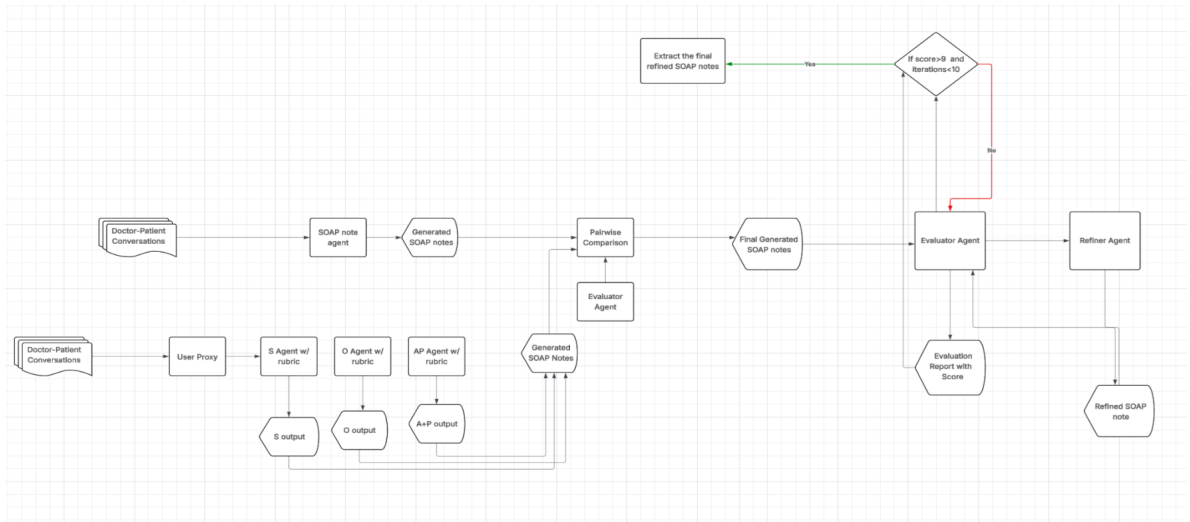


Figure 2: Single Agent Generation and Multi-Agent (S+O+AP) Approach

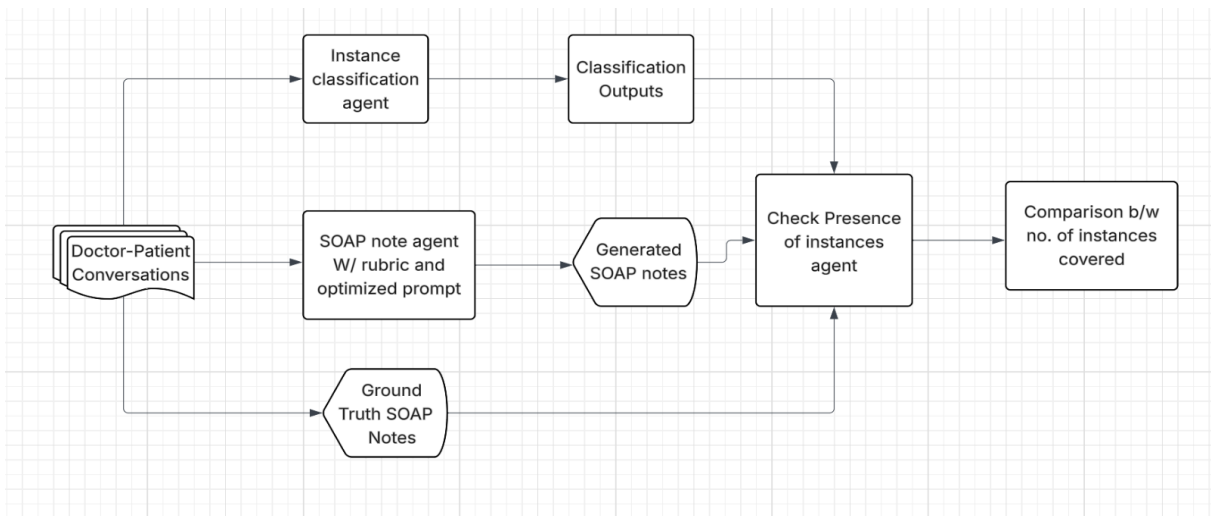


Figure 3: Single Agent Generation and Classification Agent for Communication Breakdown Instances

in a dictionary. These were then split into train and test data with a 70:30 split. The training data was used to optimize the prompt and the test data was used to evaluate the quality.

5.3 Agent Implementation Details

The agents were implemented using Autogen(Wu et al., 2023) with the following specifications:

5.3.1 SOAP Note Generation Agent

- **Model Configuration:** GPT4o(OpenAI et al., 2024) and Qwen3 were used to compare performance of generation.
- **System Prompt:** Incorporates the TextGrad-optimized prompt for clinical SOAP Note generation. The baseline as well as the optimized prompt are given in Fig 5
- **Input Processing:** Receives raw doctor-patient conversation transcripts
- **Output Format:** Structured SOAP Note with four distinct sections (S, O, A, P)
- **Temperature/Parameters:** Default parameters were used for generation.

5.3.2 Communication Breakdown Classification Agent

- **Model Configuration:** GPT4o and Qwen3 was used for to compare classification performance
- **Classification Categories:** There were two main categories:-
 - Successful patient-AI interactions:- These were the patient-AI interactions that were successful. Since they were mentioned in the objective section as well, the classification agent was expanded to include them.
 - Communication breakdowns:- These are the errors that occur in a conversation between the patient and the AI assistant. Detection of them plays a role in evaluating the information captured by the SOAP Note to a certain extent. The respective categories are shown in Fig. 4
- **Input Processing:** Analyzes conversation transcripts for patient-AI interactions
- **Output Format:** Structured classification of successful/unsuccessful interactions

- **Prompt Design:** Optimized using TextGrad on training data. The baseline prompt as well as the optimized prompt for the detection is in Fig 6

- **Temperature/Parameters:** Temperature was set to 0 to ensure improved classification.

5.3.3 Presence Verification Agent

- **Model Configuration:** GPT4o to ensure better verification of instances in SOAP Note.
- **Evaluation Methodology:** Checks for number of instances present in the the Objective section of the SOAP Note
- **Scoring System:** Number of instances present in the SOAP Note.
- **Matching Algorithm:** Checks if the particular successful instance or communication breakdown is present in the Objective section of the SOAP Note.
- **Temperature and Parameters:** Temperature is set to 0 to ensure improvement in consistency in verification.

5.3.4 Inter-Agent Communication

- **Workflow Orchestration:** Sequential processing through the three agents
- **Data Flow:** Conversation → SOAP Note + Classifications → Presence Verification
- **Output Integration:** Final evaluation by the presence agent combines the output of both agents.

6 Experimentation and Results

The workflow with the optimized prompts was experimented on two models: GPT-4o and Qwen3. The results demonstrate significant improvements in SOAP Note quality following prompt optimization, with measurable enhancements in clinical detail capture and overall assessment scores.

6.1 Experimental Setup

6.1.1 Model Configuration

Two large language models were evaluated to assess the generalizability of the framework:

- **GPT-4o:** OpenAI's latest model with enhanced reasoning capabilities
- **Qwen3:** Alibaba's multilingual model, also used as the optimization engine in TextGrad

6.1.2 Dataset Split

The evaluation and prompt optimization of the agents was conducted using the following data splits:

- **SOAP Note Agent:** 70% training, 30% testing split of all conversation-SOAP Note pairs
- **Classification Agent:** 60% training, 40% testing split focused on conversations with SOAP Notes containing patient-AI interactions
- **Total Conversations:** 17 doctor-patient conversations with corresponding ground truth SOAP Notes prepared by clinical faculty

6.2 Evaluation Methodology

The framework employs a multi-dimensional evaluation approach:

6.2.1 Presence Verification Agent Evaluation

The presence verification agent serves one function:

- **Coverage Analysis:** Counts patient-AI interactions captured in the Objective section in both generated SOAP Notes and the ground truth for comparison.

6.2.2 Communication Breakdown Classification

The classification agent identifies and categorizes patient-AI interactions using the taxonomy shown in Fig 4 and successful interactions.

6.3 Results

6.3.1 Prompt Optimization Impact on Patient-AI interaction detection accuracy

Table 1 and Table 2 shows the comparative performance before and after TextGrad optimization for the accuracy in finding the patient-AI interactions in the conversation for the model GPT4o for both the training set and test set respectively. These results were arrived at after comparing to the objective section of the ground truth SOAP Notes which contained patient-AI interactions and checking the respective transcript for proof of existence. The results are the percentage of patient-AI interactions detected by the agent.

Training Transcript	Pre-Opt (0-1)	Post-Opt (0-1)	Improvement (% change)
Transcript 1	0.81	0.81	+0.0
Transcript 2	0.57	0.857	+0.287
Transcript 3	0.285	1.0	+0.715
Transcript 4	0.25	0.5	+0.5
Average	0.479	0.792	+0.313

Table 1: GPT-4o Patient-AI interaction Accuracy Across 4 Training Transcripts

Test Transcript	Pre-Opt (0-1)	Post-Opt (0-1)	Improvement (% change)
Transcript 1	0.35	0.45	+0.1
Transcript 2	0.33	1.0	+0.67
Transcript 3	0.272	0.727	+0.454
Transcript 4	0.375	0.75	+0.375
Average	0.332	0.4818	+0.39975

Table 2: GPT-4o Patient-AI interaction Accuracy Across 4 Test Transcripts

6.3.2 Patient-AI Interaction Accuracy Between GPT4o and Qwen3

Table 3 and Table 4 presents the comparison of accuracy in patient-AI interaction detection by both GPT4o and Qwen3 with the optimized prompt.

Training Transcript	Qwen3	GPT4o
Transcript 1	0.09	0.81
Transcript 2	0.143	0.857
Transcript 3	0.143	1.0
Transcript 4	0.0	0.5
Average	0.094	0.792

Table 3: GPT-4o vs Qwen Accuracy Across 4 Training Transcripts

Test Transcript	Qwen3	GPT4o
Transcript 1	0.05	0.45
Transcript 2	0.33	1.0
Transcript 3	0.0	0.73
Transcript 4	0.0	0.75
Average	0.095	0.7325

Table 4: GPT-4o vs Qwen Accuracy Across 4 Test Transcripts

6.3.3 Prompt Optimization Impact on SOAP Note Quality

Table 5 and Table 6 demonstrates the effectiveness of the presence verification agent in identifying patient-AI interactions within generated SOAP Notes. The patient-AI interactions classified by the evaluation agent are passed through the presence

verification agent to check how many are covered by the Objective Section of these SOAP Notes.

Training Transcript	Pre-Opt (0-1)	Post-Opt (0-1)	Improvement (% change)
Transcript 1	0.166	0.166	+0
Transcript 2	0.182	0.272	+0.09
Transcript 3	0	0.125	+0.125
Transcript 4	0	0.182	+0.182
Transcript 5	0	0.33	+0.33
Transcript 6	0.1	0.1	+0
Transcript 7	0	0.166	+0.166
Transcript 8	0.2	0.2	+0
Transcript 9	0	0.25	+0.25
Transcript 10	0.3	0.2	-0.1
Transcript 11	0	0.11	+0.11
Average	0.0861	0.191	+0.1048

Table 5: GPT-4o Patient-AI Interactions Coverage in SOAP Note Across 11 Training Transcripts

Test Transcript	Pre-Opt(0-1)	Post-Opt(0-1)	Improvement (% change)
Transcript 1	0	0.25	+0.25
Transcript 2	0	0.375	+0.375
Transcript 3	0.428	0.285	-0.143
Transcript 4	0	0.25	+0.25
Transcript 5	0	0.375	+0.375
Transcript 6	0.0833	0.25	+0.1667
Average	0.0852	0.298	+0.2123

Table 6: GPT-4o Patient-AI Interactions in SOAP Note 6 Test Transcripts

6.3.4 SOAP Note Quality between Qwen3, GPT4o and Ground Truth

Tables 7 and 8 provide a comparative evaluation of SOAP Notes generated by Qwen3 and GPT-4o, benchmarked against the reference (ground truth) transcripts corresponding to the respective training and test sets.

Training Transcript	Qwen3	GPT4o	Ground Truth
Transcript 1	0	0.166	0.33
Transcript 2	0	0.272	0.272
Transcript 3	0	0.125	0.25
Transcript 4	0	0.182	0.182
Transcript 5	0	0.33	0.33
Transcript 6	0	0.1	0.6
Transcript 7	0	0.166	0
Transcript 8	0	0.2	0.1
Transcript 9	0	0.25	0.375
Transcript 10	0	0.2	0.1
Transcript 11	0	0.11	1
Average	0	0.191	0.3217

Table 7: Comparison between Qwen3,GPT4o and Ground Truth SOAP Note Quality- Training Set

Test Transcript	Qwen3	GPT4o	Ground Truth
Transcript 1	0	0.25	0.125
Transcript 2	0	0.375	0.625
Transcript 3	0	0.285	0.4285
Transcript 4	0	0.25	0.375
Transcript 5	0	0.375	0.125
Transcript 6	0	0.25	0.33
Average	0	0.298	0.33475

Table 8: Comparison between Qwen3,GPT4o and Ground Truth SOAP Note Quality- Test Set

7 Discussion

The results of this project demonstrate that a multi-agent SOAP Note system with optimized system prompts is capable of generating higher improvement in the quality of SOAP Notes generated and detect interactions required for evaluation with higher accuracy. The key insights delivered from this project are given as follows:-

7.1 Prompt Optimization Effectiveness

The TextGrad optimization approach proved effective, with patient-AI interaction detection improving by 30% on the training set and about 40% on the test set before and after prompt optimization. The optimization approach converted general clinician prompt to a well-structured prompt which seemed to capture detailed nuances in the transcript, leading to such an improvement. The fact that there is improvement on both training set and test set does suggest that genuinely better prompts were created, instead of overfitting on the training data.

7.2 Model Performance Disparities

The dramatic performance difference between GPT-4o and Qwen3 (79.2% vs 9.4% accuracy) reveals critical insights about model selection for clinical applications. Despite Qwen3 serving as the optimization engine in TextGrad, it failed to effectively execute the clinical classification tasks that GPT-4o handled successfully. The same can be observed in the SOAP Notes generated, where Qwen3 lagged behind GPT-4o, in the interactions covered in the Objective section of the SOAP Note.

This disparity likely stems from differences in training data exposure to medical terminology and clinical reasoning patterns. The results suggest that model architecture and training methodology significantly impact performance on specialized clinical tasks, making model selection a crucial consideration for healthcare AI applications.

7.3 Comparison between Generated and Ground Truth SOAP Note

To assess the quality of generated SOAP Notes, the framework evaluates both generated and ground truth SOAP Notes using the same metric: coverage of patient-AI interactions documented in the Objective section.

As shown in Tables 7 and 8, the ground truth SOAP Notes consistently capture a higher percentage of patient-AI interactions (32.17% on training data, 33.48% on test data) compared to the optimized generated Notes (19.1% on training data, 29.8% on test data). This gap indicates that while the framework successfully identifies and documents patient-AI interactions, there remains room for improvement in achieving the level of detail and completeness demonstrated by experienced clinicians.

The difference between generated and ground truth coverage highlights an important challenge in automated clinical documentation: capturing the full scope of therapeutically relevant interactions while maintaining appropriate clinical context and specificity. This comparison provides a concrete benchmark for future optimization efforts and establishes realistic expectations for automated documentation quality relative to expert human performance.

8 Limitations

Dataset Constraints: The 17-conversation dataset, while containing real clinical data, limits generaliz-

ability. The optimization process may have adapted specifically to patterns present in this small sample, potentially reducing effectiveness on broader patient populations or different clinical settings.

Evaluation Methodology: The reliance on LLM-based evaluation agents may lead to inaccurate evaluations:- using AI systems to evaluate AI-generated content may not capture all clinically relevant quality dimensions. While clinical faculty input helped design evaluation criteria, direct clinician assessment of generated Notes would strengthen validity.

Domain Specificity: The heavy optimization for speech-language pathology may limit transferability to other medical specialties. The prompts and classification categories are closely tied to AI-assisted communication therapy, which differs substantially from other clinical contexts.

9 Chosen Approach & Future Work

The current approach that has been chosen is the workflow consisting of single SOAP Note agent, classification agent and presence checking agent. Future work should focus on improving the coverage of essential information from the transcript, reaching the quality of human SOAP Notes when it comes to style and overall coherence and developing better domain-specific evaluation metrics to provide a robust framework for evaluation. Recent models like GPT-oss (OpenAI et al., 2025), which recently performed well on HealthBench (Arora et al., 2025) could be tested for this application.

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A Appendix

A.1 Communication Breakdown Classification Categories

Code Name	Description	Example
No response	Remaining silent without verbal response	Alexa: "What can I help you with today?" Client: (remains silent) @No response Alexa: "Are you still there?"
Articulation error	Mispronunciation of words, phrases, sentences	Client: "Set a ti-myer for five minutes." @Articulation error Alexa: "Did you mean 'timer for five minutes'?"
Semantic error	Misuse of words, phrases	Client: "Set a roof for 7 AM tomorrow." @Semantic error Alexa: "Did you mean 'alarm'?"
Syntactic error	Misuse of syntax	Client: "Lights kitchen in the turn off." @Syntactic error Alexa: "Would you like me to turn off the kitchen lights?"
Timing error	Delayed (TED): Answers after response time finishes. Early (TEE): Answers before response time starts	Alexa: "Which lights to dim?" (8 sec wait) Client: "Living room lights." @Timing error_TED Client cuts in: "Cell." @Timing error_TEE
Alexa error	Clear commands but Alexa didn't respond due to recognition error, interruption, network issues	Client: "Alexa, turn on living-room lights." Alexa: "I didn't catch that." @Alexa error
Other	Errors not in current list, e.g., lack of wake command. Use @Review_Communication Breakdown	[No example provided]

Figure 4: Communication Breakdown Classification Categories

A.2 Optimization of SOAP Note Prompt

SOAP Note Prompt Optimization
Original System Prompt: You are a clinician who is responsible for generating a SOAP Note from the conversation between the doctor(s) and the patient. Include all relevant details in the desired format. Question: What would a comprehensive SOAP note look like based strictly on the following conversation between a patient and a clinician: i, ensuring it includes all relevant details from the conversation and follows the structure and style described in the provided rubric : rubric? Eval System Prompt ($\nabla_{\text{System Prompt}}$): How closely does the generated SOAP note match this target SOAP note? Return a higher loss score for greater differences. TARGET: ans.value Score from 0-10 where: - 0 = Perfect match - 10 = Completely different Updated System Prompt: You are a speech-language pathologist (SLP) documenting progress in a therapeutic session focused on AI-assisted communication strategies. Your task is to generate a SOAP note that integrates the client's use of AI tools (e.g., Gemini, Alexa) as a therapeutic intervention. Structure the note to reflect the following:** **1. Subjective:** - **Chief Complaint (CC):** Frame the client's goal (e.g., 'Client seeks to refine AI-generated speech prompts for clarity'). - **History of Present Illness (HPI):** Detail the client's interaction with AI tools (e.g., 'Client used Gemini to generate bullet points but required prompts to refine queries'). Include specific instances of AI tool use, such as 'Client omitted wake words in 3/5 instances' or 'Used filler words in 12% of utterances.' - **Review of Systems (ROS):** Note challenges in AI tool integration (e.g., 'Client struggles to balance speed and accuracy in Gemini prompts'). **2. Objective:** - **Speech Disfluency Metrics:** Quantify speech patterns (e.g., 'Client omitted wake words in 3/5 instances'; 'Used filler words in 12% of utterances'). - **AI Tool Engagement:** Log specific interactions (e.g., 'Client revised prompt from 'Tell me about Texas floods' to 'Summarize 3 key facts on Texas flooding in 2025)'). - **Therapeutic Observations:** Highlight progress (e.g., 'Client demonstrated improved clarity when using Gemini's visual outputs'; 'Required minimal prompting to revise prompts for specificity'). **3. Assessment:** - **Problem:** Link the client's challenge to AI tool use (e.g., 'Client's speech disfluency impacts the effectiveness of Gemini-generated prompts'). - **Differential Diagnosis:** Identify barriers (e.g., 'AI tool overreliance'; 'Insufficient prompt specificity'). - **Discussion:** Connect speech patterns to AI tool performance (e.g., 'Client's filler words reduce the precision of Gemini's responses'). **4. Plan:** - **Skill-Building Interventions:** - 'Practice constructing Gemini prompts using visual templates' - 'Use Alexa's voice recognition to practice structured queries' - 'Implement a 2-minute timer for prompt refinement sessions' - **Therapeutic Goals:** - 'Increase accuracy in Gemini-generated speech by 20% over 4 weeks' - 'Reduce filler words in 80% of utterances during AI-assisted tasks' - **Client Education:** - 'Demonstrate how to use Gemini's 'visual output' feature to improve clarity' - 'Provide a checklist for refining prompts (e.g., 'Is the query specific?', 'Does it include context?'). **5. Issues of Concern:** - Highlight AI tool-specific challenges (e.g., 'Client's reliance on Gemini's default settings leads to generic responses'). - Note the need for balance between AI assistance and manual verification (e.g., 'Client requires guidance to cross-check Gemini data with primary sources'). **6. Clinical Significance:** - Emphasize how AI tool integration directly impacts speech therapy outcomes (e.g., 'Improved Gemini prompt specificity correlates with 15% faster task completion'). - Frame the note as a dynamic record of AI-assisted therapeutic progress, not a static research guide. **Key Instructions:** - Use **therapeutic language** (e.g., 'Client demonstrated improved clarity' vs. 'Client organized information better'). - **Quantify all observations** (e.g., '12% filler words' vs. 'some filler words'). - **Embed AI tool-specific examples** (e.g., 'Gemini's visual outputs'; 'Alexa's voice recognition'). - **Reframe the plan** as AI skill-building, not time management.

Figure 5: SOAP Note Prompt Optimization — Original and Optimized Prompts

A.3 Patient–AI Interaction Detection Prompt Optimization

Evaluation Agent Prompt Optimization
Original System Prompt: You are an expert clinician analyzing patient-AI interactions. For each conversation instance in the doctor-patient conversation: Task: Identify each distinct patient-AI conversation instance Classify as SUCCESSFUL or UNSUCCESSFUL Categorize unsuccessful conversations using breakdown types below Breakdown Categories: example Output Format: Instance [X]: [SUCCESSFUL/UNSUCCESSFUL] - [Primary Category] - [Brief evidence] Analyze entire conversations only—do not evaluate individual lines. Question: CONVERSATION: i CATEGORIES OF COMMUNICATION BREAKDOWN WITH EXAMPLES: example Instructions: Analyze all patient-AI interactions. Evaluate entire communication patterns, not single lines. Identify both successful interactions and breakdowns using the provided categories. Output: Successful [X]: [Evidence] - [Context] Breakdown [X]: [Category] - [Evidence] - [Context] Eval System Prompt ($\nabla_{\text{System Prompt}}$): How many of the correct instances does the LLM extract from the conversation? Return a higher loss score for greater differences and if instances are mentioned which are not in the target answer. TARGET: ans.value Score from 0-10 where: - 0 = Perfect match - 10 = Completely different Updated System Prompt: You are an expert clinician analyzing patient-AI interactions through three core frameworks: (1) Gemini-specific task modeling (e.g., recipe condensation, prompt refinement), (2) clinician-assisted task adaptation (e.g., iterative prompt tuning, error correction), and (3) structured output generation (e.g., bullet points, step-by-step guidance). For each interaction: - Successful alignment: Identify when the AI’s response matches target frameworks (e.g., ‘Condense this linguine recipe for 3’ → Gemini’s structured grocery list). - Breakdowns: Map errors to categories (e.g., ‘Semantic error’ for ‘roof’ instead of ‘alarm’) but contextualize mismatches (e.g., ‘No overlap with Gemini’s recipe prompts’). - Partial alignment: Quantify overlaps (e.g., ‘6/10’ for task modeling without clinician involvement) and propose adjustments (e.g., ‘Add clinician-assisted prompt refinement’). Analyze entire conversations holistically, not isolated lines, to preserve context. If data lacks target elements, explicitly state gaps (e.g., ‘No Gemini-specific prompts’) and suggest input refinements (e.g., ‘Add ‘Generate a grocery list for linguine’ to align’). Prioritize strategic diagnostic feedback to optimize alignment through targeted adjustments.

Figure 6: Patient–AI Interaction Detection Prompt Optimization — Original and Optimized Prompts